Coulomb’s law

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Relate electric force to charge magnitude and separation.

SUCCESS CRITERIA - I can...

1. Identify charge signs before predicting direction.
2. Use inverse-square scaling.
3. Calculate force with charge in coulombs and distance in meters.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “charge” and “separation” to explain your answer. Sentence starters are optional.

charge: a property responsible for electric interactions

separation: distance between the specified objects or centers

inverse square: proportional to one divided by distance squared

01 Warm-up: bring back an earlier idea

A simplified shallow-water calculation uses $v=\sqrt{(gh)}$. With $g=$**10**, $h=$**30 m** and travel distance **5 km**, find speed and travel time. **State a part of the** explanation that may not fit **what happens**.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F = k|q_1q_2|/r^2$; $k =$ **9.0×10^9 N·m²/C²**.
r is separation. Signs find attraction/repulsion; the equation gives magnitude.

Worked reasoning: Doubling the separation between two fixed charges makes Coulomb-force magnitude one quarter as large. The charge signs find attraction or repulsion separately.

Your turn: Sketch both separations for two positive charges and **label the force directions**.

Sentence starter: In my work, I used … to show … . I labeled it … .



Label charges and separation.

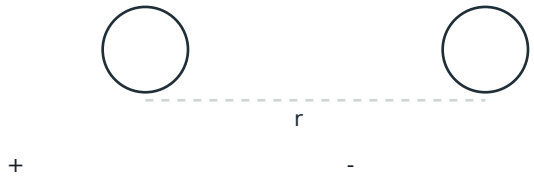
Object A Object B

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two opposite charges are moved twice as far apart. **Draw their force directions** and **compare the new force** with the original.

Sentence starter: Both … . They differ in … because … .



Each circle represents one of the labeled objects or charges; r is their center-to-center separation.

Marzano check - circle: ① ② ③

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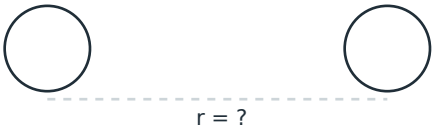
Sentence starters are optional. You may plan in any language.

04 Apply the physics

Two small charged spheres carry **+2.0 µC** and **-3.0 µC** and are **0.30 m** apart.
1. Use k = **9.0 × 10⁹ N·m²/C²** and **1 µC = 10⁻⁶ C**.
2. **Label the diagram**, calculate force magnitude and **draw the force** on each sphere.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Charge 1 Charge 2

Marzano check - circle: 1 2 3

05 Explain your reasoning

One charge has twice the magnitude of the other. Does it exert twice the force on the smaller charge? Explain.

Word help: magnitude = size of a quantity without its direction

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **600 Hz** tone lasts **0.4 s**. **Find number of cycles** and period.
Label a wave sketch to **support the calculation or comparison**.

Sentence starter: I found … [unit]. This value means … in this situation.



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

Both charge magnitudes double and distance triples. Express the new force as a fraction of the old.

Sentence starter: I notice … . This supports … because … .



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two small spheres carry **+1.0 µC** and **+4.0 µC** and are **0.20 m** apart. Use k = **9.0 × 10⁹ N·m²/C²**. **Calculate the force magnitude** and **draw both force arrows**. Then triple the separation and recalculate.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found … [unit]. This value means … in this situation.



Label charges and separation.

Object A Object B

Marzano check - circle: 1 2 3